

Extra: funció que fa recompte

```
DECLARE
FUNCTION GET_EXEMPLARS_PER_GENERE(p_genere VARCHAR2) RETURN NUMBER IS
    TYPE t_ids IS TABLE OF LLIBRE.ID%TYPE;
    v_ids t_ids;
    vExemplars NUMBER := 0;
    vAux NUMBER;
    vIDX NUMBER;
BEGIN
    SELECT count(*) INTO vAux FROM GENERE WHERE NOM = p_genere;
    IF vAux = 0 THEN
        DBMS_OUTPUT.PUT_LINE('Gènere desconegut');
        RETURN 0;
    END IF;
    SELECT ID_LLIBRE BULK COLLECT INTO v_ids FROM LLIBRE_GENERE WHERE NOM_GENERE = p_genere;
    vIDX := v_ids.FIRST;
    WHILE vIDX IS NOT NULL LOOP
        SELECT EXEMPLARS INTO vAux FROM LLIBRE WHERE ID = v_ids(vIDX);
        vExemplars := vExemplars + vAux;
        vIDX := v_ids.NEXT(vIDX);
    END LOOP;
    RETURN vExemplars;
END;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Hi ha '||GET_EXEMPLARS_PER_GENERE('Poesia')||' exemplars del gènere Poesia.');
```

Extra: funció que reb una nested table

```
DECLARE
  TYPE t_genres IS TABLE OF VARCHAR2(50);
  vGenres t_genres := t_genres('Poesia', 'Astronomia', 'Fantasia');

  FUNCTION GET_EXEMPLARS_PER_GENERES(p_genres t_genres) RETURN NUMBER IS
    vExemplars NUMBER := 0;
    vAux NUMBER;
    vIDX NUMBER;
  BEGIN
    vIDX := p_genres.FIRST;
    WHILE vIDX IS NOT NULL LOOP
      SELECT SUM(EXEMPLARS) INTO vAux FROM LLIBRE
        JOIN LLIBRE_GENERE ON LLIBRE_GENERE.ID_LLIBRE = LLIBRE.ID
        WHERE LLIBRE_GENERE.NOM_GENERE = p_genres(vIDX);
      IF vAux IS NOT NULL THEN
        vExemplars := vExemplars + vAux;
      END IF;
      vIDX := p_genres.NEXT(vIDX);
    END LOOP;
    RETURN vExemplars;
  END;
BEGIN
  DBMS_OUTPUT.PUT_LINE('Hi ha ' || GET_EXEMPLARS_PER_GENERES(vGenres) || ' exemplars dels gèneres indicats.');
```

Extra: Procedure standalone



```
CREATE OR REPLACE PROCEDURE UPDATE_PRODUCT_PRICE(p_id_product number, p_price number)
IS
BEGIN
    IF (p_price < 0 or p_price > 10000) THEN
        DBMS_OUTPUT.PUT_LINE('Preu invàlid');
    ELSE
        UPDATE PRODUCTS SET list_price = p_price WHERE product_id = p_id_product;
    END;
END;
```

U8: soluciones

Exercici 1: solució

```
DECLARE
    v_countryName varchar2(5) := 'Spain';
    v_countryID varchar2(2) := 'ES';
    v_regionID number;
BEGIN
    SELECT region_id into v_regionID FROM regions WHERE region_name = 'Europe';
    INSERT INTO COUNTRIES VALUES (v_countryID, v_countryName, v_regionID);
    DBMS_OUTPUT.PUT_LINE('v_countryName: ' || v_countryName || ', v_countryID: ' || v_countryID || ',
        v_regionID: ' || v_regionID);
END;
```

- També es pot utilitzar %TYPE:

```
v_countryName COUNTRIES.COUNTRY_NAME%TYPE := 'Spain';
v_CountryID COUNTRIES.COUNTRY_ID%TYPE := 'ES';
v_regionID COUNTRIES.REGION_ID%TYPE;
```

Associative arrays: sol·lució exercici



```
DECLARE
  type t_sales is table of pls_integer index by varchar2(10);
  c_sales t_sales;
BEGIN
  SELECT count(*) into c_sales('Blake') from orders join employees on salesman_id = employee_id
    where first_name = 'Blake' and status = 'Shipped';
  SELECT count(*) into c_sales('Daisy') from orders join employees on salesman_id = employee_id
    where first_name = 'Daisy' and status = 'Shipped';
  SELECT count(*) into c_sales('Lily') from orders join employees on salesman_id = employee_id
    where first_name = 'Lily' and status = 'Shipped';

  DBMS_OUTPUT.PUT_LINE('First: ' || c_sales.FIRST || ', ' || c_sales(c_sales.FIRST));
  DBMS_OUTPUT.PUT_LINE('Second: ' || c_sales.NEXT(c_sales.FIRST) || ', ' || c_sales(c_sales.NEXT(c_sales.FIRST)));
  DBMS_OUTPUT.PUT_LINE('Last: ' || c_sales.LAST || ', ' || c_sales(c_sales.LAST));
END;
```

Varrays: solució exercici

```
DECLARE
  TYPE t_podium is varray(3) of varchar2(50);
  c_podium t_podium;
BEGIN
  SELECT name bulk collect into c_podium from (
    select first_name||last_name as name, count(distinct order_id) as orders from employees
    join orders ord on ord.salesman_id = employee_id and ord.status = 'Shipped'
    group by first_name||last_name
    order by count(distinct order_id) desc)
  where rownum <=3;

  DBMS_OUTPUT.PUT_LINE('1st: '||c_podium(1));
  DBMS_OUTPUT.PUT_LINE('2nd: '||c_podium(2));
  DBMS_OUTPUT.PUT_LINE('3rd: '||c_podium(3));

  c_podium.TRIM(2);
  DBMS_OUTPUT.PUT_LINE('1st: '||c_podium(1));
END;
```

Exercici: solució



```
DECLARE
    v_product PRODUCTS%ROWTYPE;
BEGIN
    SELECT * INTO v_product FROM products WHERE product_name = 'G.Skill Ripjaws V Series' and
        description like '%DDR4-3333%' and rownum = 1;
    v_product.description := replace(v_product.description, 'DDR4-3333', 'DDR4-3600');
    v_product.standard_cost := 697.56;
    v_product.list_price := 779.99;

    INSERT INTO products(product_name, description, standard_cost, list_price, category_id) VALUES
        (v_product.product_name, v_product.description, v_product.standard_cost,
        v_product.list_price, v_product.category_id);

END;
```


Records amb col·leccions: solució exercici

```
DECLARE
    TYPE t_varr IS VARRAY(3) OF LLIBRE%ROWTYPE;
    v_arrllibres t_varr;
BEGIN
    SELECT * BULK COLLECT INTO v_arrllibres FROM LLIBRE ORDER BY EXEMPLARS DESC FETCH FIRST 3 ROWS ONLY;
    DBMS_OUTPUT.PUT_LINE('1: ' || v_arrllibres(1).titol);
    DBMS_OUTPUT.PUT_LINE('2: ' || v_arrllibres(2).titol);
    DBMS_OUTPUT.PUT_LINE('3: ' || v_arrllibres(3).titol);
END;
```